DNA MODIFICATION FRAMEWORK

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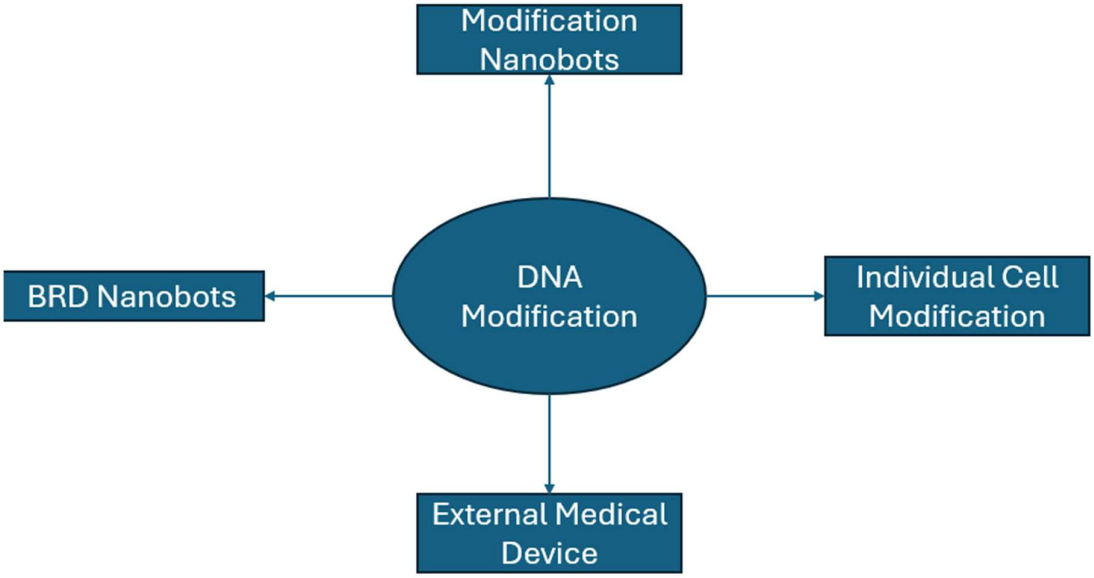
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Introduction

This DNA modification framework is designed to allow for researchers to modify DNA as safely and reliably as possible. This framework is also designed to show the uses of nanotechnology within DNA. This framework was developed as DNA is a source of multiple diseases as well as a critical and essential component needed for reproduction and growth within a human being. This framework focuses on modifying DNA within each cell using nanobot to remove genetic diseases as well as improve the genetics of humans overall.

This framework cannot promise perfect safety, reliability or modification as no system or solution is able to. However, this system promises to try its best to provide a reliable and safe solution to modify DNA on a scale that humanity has not seen yet. This framework is targeted towards researchers and nano engineers. However, those studying medical technology or are just interested in this topic are more than welcome to read the content within this framework. This framework is designed to allow researchers to fight against genetic diseases, research DNA at a deeper level and improve human genetics in the long-term. Make sure to adapt the components and the designs used in this framework to real-world test results and usage.



In the spider diagram above, we can see the 4 main concepts involved within this DNA modification framework. In the diagram, we can tell that there will be 2 nanobots involved with different functions as well as an external medical device. We can also see that the goal with the nanobots is to modify thousands if not millions of individual cells instead of whole genomes at once.

Nanobot Deployment Section

- **This nanobot deployment section has 5 steps just like the other medical frameworks I have created.**
- **This section is important because the nanobot needs to deploy safely and in a specific way to perform its job correctly.**
- **The first step is nanobot training**
- **The nanobots must be trained to target a specific type of cell and move through it to ensure that it is functioning correctly.**
- **This can be done by testing the nanobot on dead cells, moving through certain fluids and configuring the ASIC.**
- **This step is important to ensure that the nanobots know their mission before they proceed.**
- **The second step is nanobot verification.**
- **The nanobots must be checked for any major issues before deployment to ensure that there are no issues before it even enters the body.**
- **This can be done by checking the physical components for any damage and running diagnostic scans to ensure that the logic is working correctly.**
- **This step is important to ensure that the nanobots are truly ready for deployment.**
- **The third step is nanobot deployment**
- **The nanobots must be deployed into the patient's body by medical professionals to ensure that they can do their job.**
- **This can be done by injecting them into the patient's arm specifically on a vein for easier blood access.**
- **This step is important to allow the nanobots to enter the human body directly where they can begin work immediately.**
- **The fourth step is nanobot functions**
- **This is where the nanobots will perform their DNA modification and repairing of the cell after it's been modified.**
- **This can be done by letting them do their job, monitoring via the external medical devices and monitoring the patient for any signs of side-effects.**
- **This step is important as it is where the actual DNA modification work is performed and where the outcome is produced.**
- **The fifth step is nanobot disposal**
- **This is where the nanobots will dissolve harmless into the body's bloodstream where it can be dealt with by the digestive system.**
- **This can be done by letting the nanobots dissolve by themselves through their own logic or remotely giving the command to do so.**
- **This step is important to ensure that the nanobots are removed safely after usage to prevent causing blockages or having the risk of modifying modified cells.**
- **Make sure that the patient is monitored for any effects of damage.**

Breach, Repair and Deliver (BRD) Nanobots Design

- **This type of nanobot is designed to breach into the cell, deliver the smaller nanobot and repair the breach on the cell.**
- **I will be discussing the specifications of this design physically and what systems it requires digitally.**
- **I will begin by discussing the physical components and specifications**
- **For size, this nanobot will be 4500 nanometres to accommodate the various components and systems it will include.**
- **This is because this nanobot will include the normal components as well as additional like mechanical movement and delivery systems.**
- **This size may increase to 5000 nanometres to provide more space as this nanobot includes more components compared to most of my nanobots.**
- **For power source, the nanobot will use a solid-state thin battery which is the usual battery used for the nanobots.**
- **For processing, the nanobot will use ASIC as usual since a general-purpose CPU or microprocessor will never be able to fit.**
- **For storage, the nanobot will use RRAM to store any data that it will have during its functions.**
- **For movement, the nanobot will mainly travel using the blood however it will also include tiny mechanical legs.**
- **These mechanical legs are needed to allow the larger nanobot to position itself perfectly to break through the nucleus envelope and launch the smaller nanobot.**
- **For the delivery mechanism, the smaller nanobot will be inside of a hangar that is big enough to fit it with a door that can be opened to allow it to exit it.**
- **The smaller nanobot will be turned off until the larger nanobot gives it an electrical signal to turn on before the doors open.**
- **For the repair system, the nanobot will have a box/container where it can release Annexins to repair the pores.**
- **For the transmitter, the nanobot will use Zinc oxide (ZnO) thin-film piezoelectric for the actual transmitter to ensure it can communicate via ultrasound signals.**
- **I will now moving on to discuss the logical systems that are involved for this nanobot.**
- **For movement, the logical system will only use the legs once it has breached the cell membrane to get to the nucleus.**
- **For targeting, the nanobot will use electrical bursts of 0.8V to breach through the two layers of the cell to get to the nucleus.**
- **This is to ensure that the nanobot doesn't kill the cell but to make the pore big enough for the nanobot to fit.**
- **For communication, the nanobot will transmit data on the destruction It does and the status of launch and other simple performance metrics.**
- **For deployment of the smaller nanobot, the BRD nanobot will only deploy this nanobot once the second pore has been created to ensure precision.**

Modification Nanobot Design

- **This nanobot is designed to modify the DNA of the cell as its one and only function since it is too small to do anything else.**
- **I will be discussing 3 areas of this design.**
- **The first area is physical components**
- **For size, this nanobot will be 2500 nanometres as there isn't a very high number of components involved like there is for the BRD nanobot.**
- **This size was chosen mainly as this is a very specialised nanobot fit for one purpose only.**
- **For processing, the nanobot will use ASIC since it is only doing one function and it is perfect for that.**
- **For storage, the nanobot will use RRAM since it doesn't need to store a high amount of data and its one of the only data options at that scale.**
- **For movement, this nanobot will not have any mechanical legs since the distance will be so close.**
- **This was chosen because the nanobot can rely on the flow to move and bind to the chromosomes when it is closed.**
- **For modification, the nanobot will store CasMINL to release it when it binds to the chromosomes within the nucleus.**
- **CasMINL was chosen due to its small size and its ability to modify DNA well which is needed for such a specialised and small nanobot.**
- **For binding, the nanobot will use DNA hybridization to make sure it stays on the chromosomes during the release.**
- **For the transmitter, Ultrasound will be used with the material of Zinc oxide (ZnO), thin-film piezoelectric for the actual transmitter.**
- **For targeting, the nanobots will use the DNA hybridization sensors to ensure they can find the chromosomes that they need**
- **I will now discuss the logical systems this nanobot will have.**
- **For communication, this nanobot will confirm if the DNA has been modified and give data on the DNA that are possible with its sensors.**
- **This is an important system as the researchers need to know if the outcome was a success and collect data on DNA.**
- **For modification, the nanobot will release the CasMINL when it is to bind to the chromosome.**
- **This is important because the nanobot needs to make the release of the CasMINL as precise and accurate as possible for reliability.**
- **For disposal, the nanobot will transmit any post-destruction data and initiate self-destruct protocols**
- **This is important to ensure that the nanobot doesn't cause any damage from long-term durations of staying within the nucleus of a cell**
- **This design can be adapted to use different DNA modification tools if it is needed.**

Nanobot External Medical Device Design

- **This external medical device is designed to provide conflict resolution, a database and coordination between the nanobots.**
- **This external medical device is needed to ensure that the nanobots don't modify already modified cells and to ensure that data is received and can be reviewed.**
- **I have decided to name this medical device which is the nanobot conflict resolution, coordination and data storage interface (NCRCDSI).**
- **I will begin by discussing the physical components for this external medical device.**
- **For the power source, this device will use a wall outlet to receive power from the main electrical generator.**
- **As a secondary backup, this device could use a battery that can last for hours in case of a power outage during a live nanobot deployment.**
- **This system was chosen as this device needs to be resilient especially when working with nanobots and the potential danger they can have uncoordinated.**
- **For the processing, the device can use a general-purpose CPU to perform the complex functions of conflict-resolution and coordination.**
- **The CPU that can be specifically used is ARM Cortex-A which is already used and proven in multiple medical devices.**
- **This approach was chosen as an embedded system wouldn't have been powerful enough to coordinate and solve conflicts in real-time at the scale of hundreds to thousands of nanobots.**
- **For storage, the device can use an internal drive with about 256GB of space to be able to fit in an operating system, the database and other needed data.**
- **The device can specifically use SSD which is extremely fast and can support the needed capacity that is needed for an operation such as this.**
- **This storage system was chosen as other alternatives either had too low capacity or not the needed processing speed.**
- **For the transmitter, the device will use piezoelectric transducers to check for ultrasound signals as much as possible and reliably**
- **For range, this external device is designed to only handle one patient at a time since it would be too complex at first to handle multiple patients at once.**
- **I will now discuss the logical systems in this device.**
- **For the operating system, it will be Linux-based but customised to perform the real-time conflict resolution, coordination, data management as well as a GUI.**
- **For the conflict resolution, the device logs what each cell each nanobot has modified using serial numbers to arrange it.**
- **Once the data has been received, the device uses this data to send signals to avoid the nanobots to work from already modified cells.**
- **For nanobot failure, the device tries to establish contact 3 times before assuming failure. It does not assign a new nanobot in case of damage to the cell or silent success.**

DNA Nanobot Modification Workflow

- **This modification workflow is designed to show the whole process from pre-deployment to post-deployment.**
- **First, the nanobot are trained by configuring their targets, loading the DNA modification chemicals and testing them on extracted cells.**
- **Second, the nanobot are verified for functionality by running diagnostics and checking for physical damage on the components.**
- **Third, the nanobots are deployed into the patient's body via their arm directly on one of their veins.**
- **Fourth, the nanobots travel through the blood to reach cells all through the body to ensure a larger coverage.**
- **The nanobots specifically travel through capillaries, arteries and veins to reach the allowed cells in their list.**
- **Fifth, the BRD nanobots penetrate the cell membrane through a controlled electrical impulse creating a pore for it to fit through.**
- **Sixth, the BRD nanobots transmit that they have entered to the cell to the external medical device.**
- **Seventh, the external medical device starts collecting real-time data, performing conflict resolution and coordinating nanobots to avoid remodification.**
- **Eighth, the BRD nanobots use their mechanical legs to move to the nucleus envelope detecting it using their sensors**
- **Ninth, the BRD nanobots breach the envelope by using another electrical impulse to create a pore to allow for the modification nanobots to enter.**
- **Tenth, the modification bots are activated inside their hangar, and the doors are opened to launch them inside the nucleus.**
- **Eveleth, the modification bots enter the nucleus successfully through the pore and begin locating the chromosomes using their sensors.**
- **Twelfth, the modification bots locate the chromosomes and bind to them preparing to release their stored CasMINL.**
- **Thirteenth, the modification bots release the CasMINL into the chromosome starting the modification process.**
- **Fourteenth, the BRD nanobots finish repairing the pore on the nucleus envelope using their Annexins.**
- **Fifteenth, the CasMINL successfully modifies the chromosomes meaning that the DNA was changed within the cell successfully.**
- **Sixteenth, the BRD nanobot finishes repairing the pore on the cell membrane and transmits post-repair data.**
- **Seventeenth, The BRD nanobot and the modification nanobot dissolve as their task has been completed successfully.**
- **Finally, the researchers transfer the data from the NCRCDSI for analysis in the lab using workstations.**

DNA Modification Framework Trade-offs

- **This DNA modification framework has trade-offs just like any other system I have created.**
- **It is important that while this framework is grounded within real biology and nanotechnology it has not fully been proven.**
- **I have identified 5 main trade-offs with this framework.**
- **The first trade-off is unperfected modification.**
- **This trade-off exists as perfect modification can never be achieved within all cells that contain DNA.**
- **This means that there will always be some failure in DNA modification as nothing is perfect after all.**
- **This trade-off can be managed by using precise modification, careful delivery mechanisms and accepting that it won't be perfect.**
- **The second trade-off is non-complete coverage.**
- **This trade-off exists as nanobots cannot modify every single cell that contain DNA within the human body.**
- **This is because there are trillions of cells meaning they would need to be trillions of nanobots to modify every DNA-containing cell.**
- **This trade-off can be managed by focusing on the most critical cells like reproduction cells which will have a bigger impact than most.**
- **The third trade-off is complexity.**
- **This trade-off exists as this modification process can be quite difficult to understand and implement.**
- **This means that there may be mistakes or misunderstandings during real-life modifications using this framework.**
- **This trade-off can be managed by training staff, having medical contingencies for worst-case situations and ensuring that the process is clear.**
- **The fourth trade-off is time-consuming**
- **This trade-off exists as this modification process can take a long time since it has a large coverage and lots of verification processes inside and outside of the body.**
- **This means that DNA modification using this solution won't be a quick process as most solutions in medicine aren't.**
- **This trade-off can be managed by automating processes where possible, making nanobots faster and more efficient and improving DNA modification.**
- **The fifth trade-off is nanobot failure**
- **This trade-off exists as these nanobots aren't perfect meaning that failure is always a risk even in the best of designs.**
- **This means that nanobots may fail mid process during the modification which could be major.**
- **This trade-off can be managed by creating safeguards against failure, creating systems that isolate failures and constantly monitoring the patient.**